Project 1: A numerical method for a dynamical system

In this project, you will work with a small group of peers to select an engineering-relevant dynamical system, decide on an appropriate numerical method to solve that dynamical system, and perform systematic studies that let you explore something that excites you about the dynamical system that you've chosen.

Your project grade will be based on an evaluation of a technical report that you submit. Your report should be typeset in LaTeX, and your submission should be the `.pdf` output of that document. The grade will be based on the following components. You have freedom in what report structure you choose, but may wish to adopt a report structure that mirrors the evaluation criteria below.

Evaluation

1. The premise — 22 pts

Create an AE-relevant technical premise that excites you. Specifically, you should develop a prompt or question/set of questions that you are excited to answer. You have significant autonomy in what premise you pursue, but make sure that resolving that premise/addressing the questions you pose requires *numerically* solving an *initial value problem*. Start by framing the premise or question that motivates you, and explain why addressing it will require studying an initial value problem. Present the initial value problem clearly, as it pertains to your premise, and explain why numerical methods are needed to address your prompt (for example—if the initial value problem has an analytical solution, it is unlikely you have chosen the right premise). Be sure that your premise is conveyed clearly in words as well as any necessary mathematics to define the initial value problem and the problem parameters that your investigation depends on.

2. The method — 22 pts

Present a numerical method that is appropriate to study the questions that you've asked of your dynamical system. Your presentation will be evaluated based on (i) your justification for why the method is appropriate to address the questions you want to probe in your dynamical system, using accuracy, stability, and cost considerations; (ii) your mathematical derivation of the method including its error and stability properties; (iii) an algorithmic summary of how the method advances a solution from some time instance t_k to t_{k+1}.

3. Correct implementation — 22 pts

Demonstrate a correct implementation of the method. Implement the method yourselves, from scratch: do not call an outside time integrator such as `scipy.integrate.solve_ivp` to advance the solution. (If you choose to use an implicit numerical: you can, as usual, call an outside solver such as `scipy.optimize.fsolve` for the nonlinear algebraic equations that the implicit treatment yields at each time step.) Your full code is part of the submission (see the Reproducibility section below), so be sure to adhere to this requirement. The demonstration should have two elements:

(i) Error studies that show your method converges at the expected rate, *and* that it converges to the solution of the correct equations. A convergence study against a solution computed with a very small Δt confirms the rate, but not that the answer is right: a mistake in the equations converges just as cleanly. To confirm the answer, compare against something you know independently—for example, an exact solution of a special case of your problem (a limit in which the IVP becomes linear, or in which a term drops out), a conservation law your system must satisfy, or published data.

(ii) An evaluation of which simulation parameters (e.g., time step size) your method needs to use so that you can accurately study the problem of interest.

4. Results — 22 pts

Present results that meaningfully address the question you ask. The results should utilize professional, concise text along with clear companion figures that get at the heart of the questions you pose in part 1. Be thoughtful about these results! Raw results that plot the trajectory of your state variables as a function of time are rarely the most direct way of addressing the specific questions you posed.

5. Contributions — 6 pts

A summary description of everyone's roles in the project. These summary descriptions should provide a rough estimate of the percentage of the project that each group member contributed. There is flexibility in how the group structure is set up. Everyone can work equally on every part of the project, or you can provide team leads for things like defining the problem statement, picking and justifying the method, coding the method, etc. A few essential elements of the group contribution plan:

(i) Each student must contribute meaningfully to at least one of the rubric items (1)–(4). That is, it is insufficient to contribute nothing to rubric items (1)–(4), but to try to "offload" this by coordinating everyone (on Discord, say) and being responsible for the final group writeup. Those latter elements are important and worth mentioning in the summary descriptions, but are insufficient unto themselves for full credit on the group project. Insufficient demonstration of contribution to at least one of the rubric items (1)–(4) could lead to point deductions from the nominal group grade.

(ii) The estimated percentage of the member contribution to the project must be meaningful. It is not essential that everyone's percentages are identical, but noticeable discrepancies in percentages could lead to point deductions from the nominal group grade.

6. Reproducibility — 6 pts

You should provide a link to your code that contains the full version of the code, including whatever is needed to run the convergence tests and generate any results that you use in your report. For full credit, the code must be well organized, ideally distinguishing between the code that implements the numerical method (perhaps as a function that time marches the solution), and the other codes that perform convergence tests and generate results. **Code that is hosted as a well-constructed repository on GitHub can be awarded up to 3 bonus points on the group project grade.**

Total: 100 points, plus up to 3 bonus points

Weight. Remember that Project 1 (the individual and group portions combined) is worth 30% of your semester grade. That is, this group project portion is worth 15% of your semester grade. As such, I am looking for you to go well beyond what is expected from a homework submission, and take ownership of your work on this project. Going "beyond what is expected" will mean different things to different people. I want to build in flexibility for that "above and beyond" component to be driven by something you're passionate about. Just be sure to work hard and have fun!

Generative AI. As always, use of generative AI is welcome. If you use generative AI, be sure to give an overview of how you used it in an Acknowledgments section. The overview need not be detailed, but it should give a clear picture of how generative AI was used throughout the project, from ideation to method selection and justification, to implementing and testing the code, to generating figures and results.